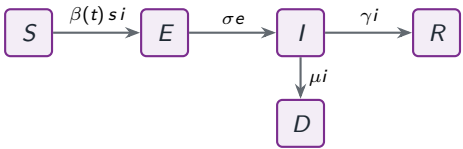
Why transmission must change over time

The problem

- Ebola spreads by contact with body fluids, and with the dead at funerals
- isolation, safe burials and contact tracing cut transmission as the epidemic went on: β is **not** a constant
- surveillance reports only cases and deaths; S and E are never measured



$\dot{s} = -\beta s i, \quad \dot{e} = \beta s i - \sigma e, \quad \dot{i} = \sigma e - (\gamma + \mu) i, \quad \dot{r} = \gamma i, \quad \dot{d} = \mu i$

fractions of N, so $s + e + i + r + d = 1$ at all times

	Meaning	Value
N	population (2014 census)	7,092,113
σ	incubation rate (9.4 days)	0.1064 /day
γ	recovery rate (7 days)	0.1429 /day
μ	death rate	0.1000 /day
β	transmission rate (baseline)	0.4858 /day
CFR	$\mu/(\gamma + \mu)$	41.2%
R_0	$\beta/(\gamma + \mu)$	2.00

Assumptions

- closed population, homogeneous mixing
- recovered people stay immune
- σ, γ fixed from WHO clinical data, so the data only have to determine $\beta(t)$ and μ

Published models: how others handled $\beta(t)$

Module 1 ▪ each makes one assumption about how transmission changes, and each assumption has a cost

Chowell et al. (2004) Congo 1995, Uganda 2000

- SEIR model; β is high until control begins, then falls
- fitted to case counts: $R_0 \approx 1.83$ (Congo), 1.34 (Uganda)
- × the **shape** of the fall is chosen in advance

Legrand et al. (2007) three routes of infection

$$\lambda(t) = \beta_I \frac{I}{N} + \beta_H \frac{H}{N} + \beta_F \frac{F}{N}$$

- community (I), hospital (H) and funeral (F) transmission
- × H and F follow I, so from one national case curve the three β's trade off: **many sets fit equally well**

Rivers et al. (2014) a policy step

$$\beta(t) = \beta_0 \left[1 - \delta \, \mathbf{1}(t \geq t_{\text{policy}})\right]$$

- β drops by a fraction δ on a chosen date ($\mathbf{1}$ switches from 0 to 1 there)
- × an **instant jump** at an **assumed date**; real behaviour changes over weeks

Althaus (2014) exponential decay

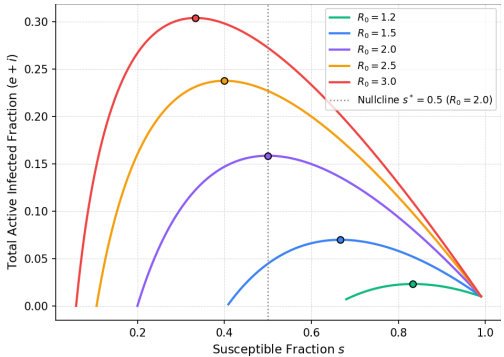
$$\beta(t) = \beta_0 \, e^{-k(t-t_0)}, \qquad \frac{dR_t}{dt} = -\frac{\beta s}{\gamma + \mu}\left[k + \beta i\right] < 0$$

- smooth, but R_t can **only fall**
- × once $R_t < 1$ it never rises again: **no resurgence**, no relaxation of control

The gap: every published model fixes the *shape* of $\beta(t)$ in advance (a date, a step, a decay). Our PINN learns $\beta(t)$ from cases and deaths, with no date or shape assumed.

Phase plane: where the epidemic peaks

Module 1 ▪ exact formulas, checked against an accurate ODE solver (DOP853)



$R_0 = 1.2$–3.0; time runs right to left; dots = predicted peaks

Divide the equations: i cancels exactly

$$\frac{d(e+i)}{ds} = -1 + \frac{1}{R_0\, s}$$

Peak where the slope is zero: $s^* = 1/R_0$

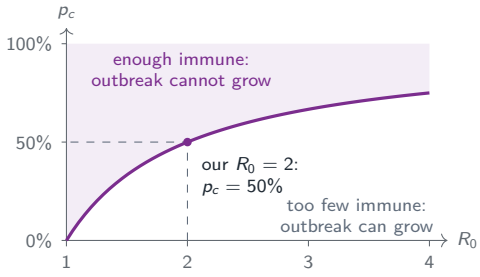
$$(e+i)_{\max} = (e_0 + i_0) + s_0 - \tfrac{1}{R_0}\left[1 + \ln(R_0 s_0)\right]$$

- $R_0 = 2$: peak **15.85%** at $s = 0.5$ (i alone about 4.8%)
- formula vs. ODE: relative error at most 1.3×10^{-5}
- right of s^*: $e + i$ grows ($R_t > 1$); left of s^*: it shrinks ($R_t < 1$)

Why it matters: the peak comes exactly when the susceptible share falls to $1/R_0$, the herd-immunity threshold (next slide). These exact formulas are also the checks used for the solvers (Module 3) and the PINN (Module 4).

Herd immunity and the control threshold

Module 1 ▪ how many must be immune, or how much transmission must be cut, before an outbreak cannot grow



R_0	1.5	2	3	4
$p_c = 1 - 1/R_0$	33%	**50%**	67%	75%

Where it comes from

- one case infects $R_t = R_0\, s$ people (only susceptible contacts count), so growth stops once $s < 1/R_0$: the immune share must exceed $p_c = 1 - 1/R_0$
- same threshold as Module 2: a disease-free state is stable only if $R_0 s^* < 1$

Two ways to reach it

- **immunity**, by vaccination or past infection; there was no licensed Ebola vaccine in 2014–15
- **cutting transmission** by δ: $R_t = (1-\delta)R_0\, s < 1$ when $\delta > \delta_c = 1 - 1/R_0$ (50% for $R_0 = 2$)

Two subtleties

- **overshoot:** reaching p_c only marks the *peak*; those still infectious keep infecting, so with $R_0 = 2$ about 80% are infected in the end, not 50%
- **timing:** later on less is needed, $1 - 1/R_t$: 37.5% at $s = 0.8$

What it means for control: without a vaccine, isolation, safe burials and tracing together must cut transmission by **more than 50%** at the start, and the earlier they act, the smaller the epidemic.